Disorders of Renal Function

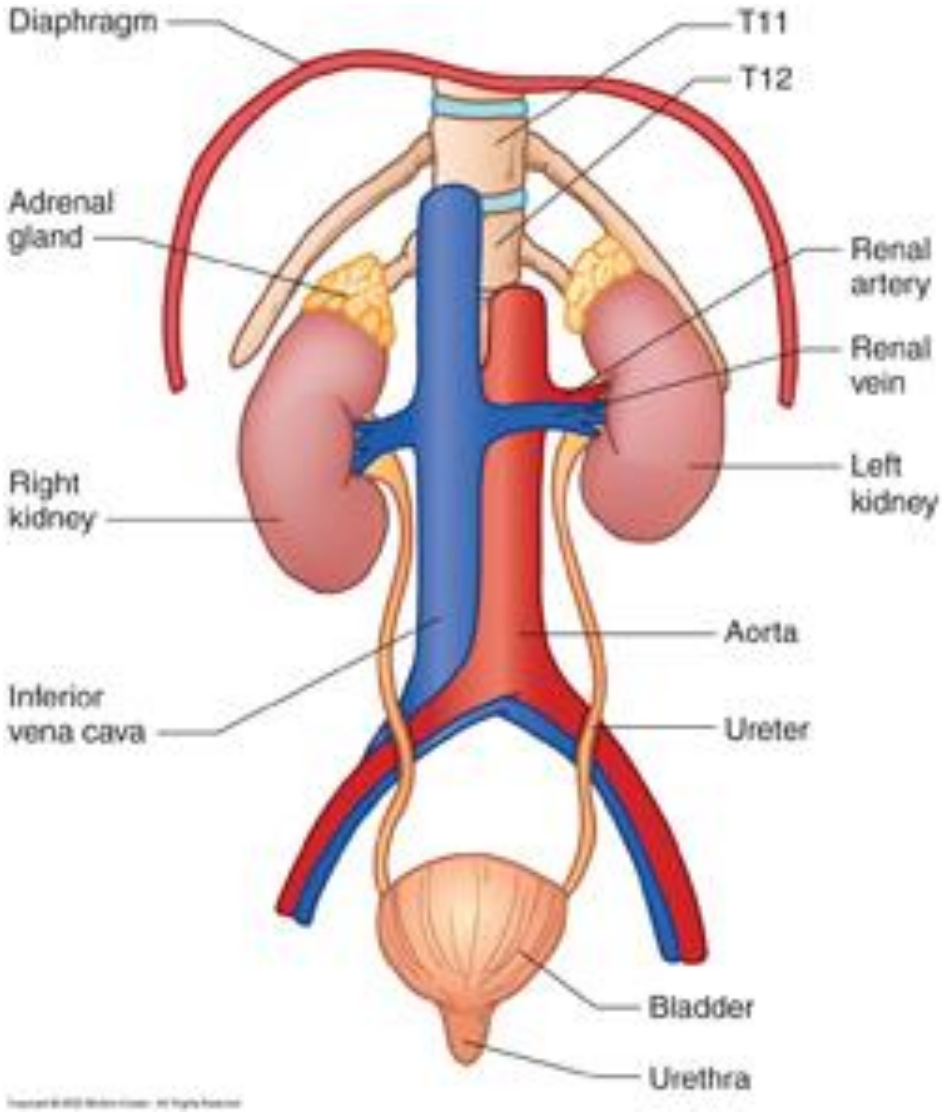
Yasir Eltahir

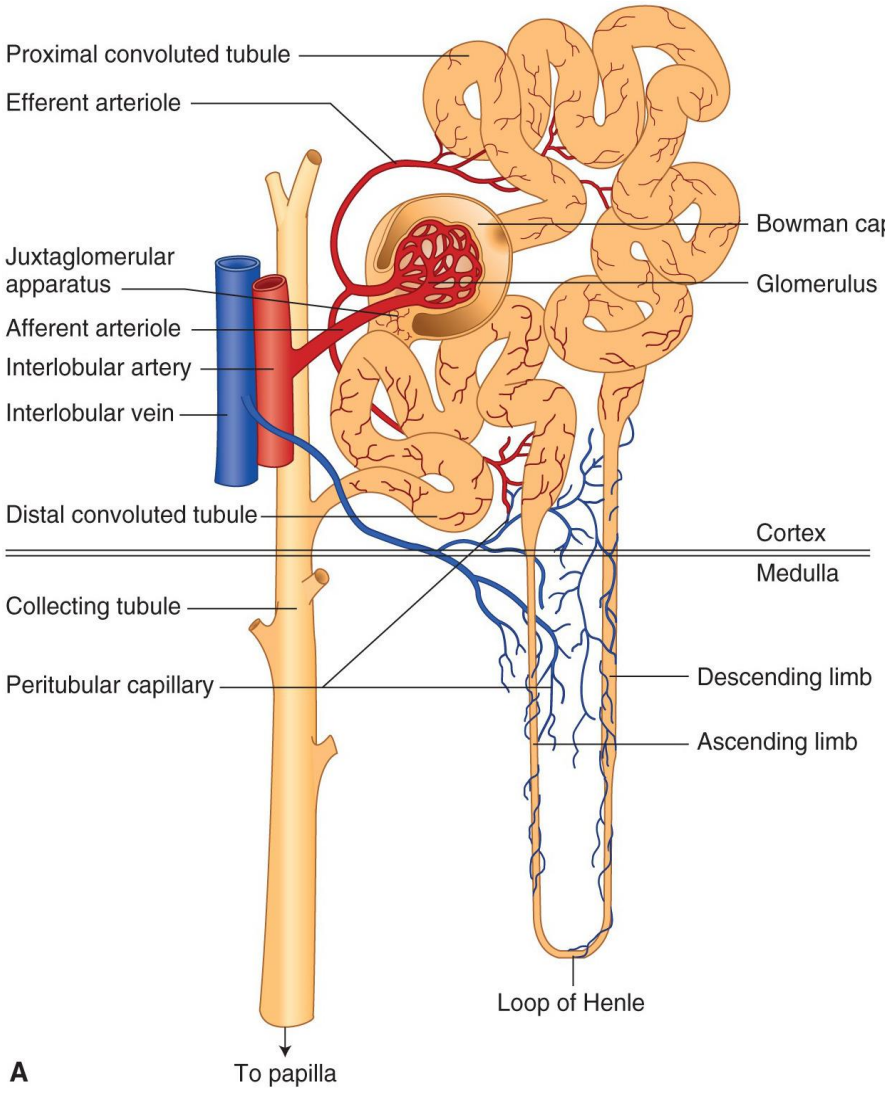
Objectives

- Explain **developmental defects** of the renal system.
- Identify the pathophysiology, risk factors, and clinical manifestations of **UTIs**.
- Differentiate between **nephrotic and nephritic** syndromes.
- Understand the formation of **renal stones** and their complications.
- Describe Wilms **tumor** in children, its presentation, and management.

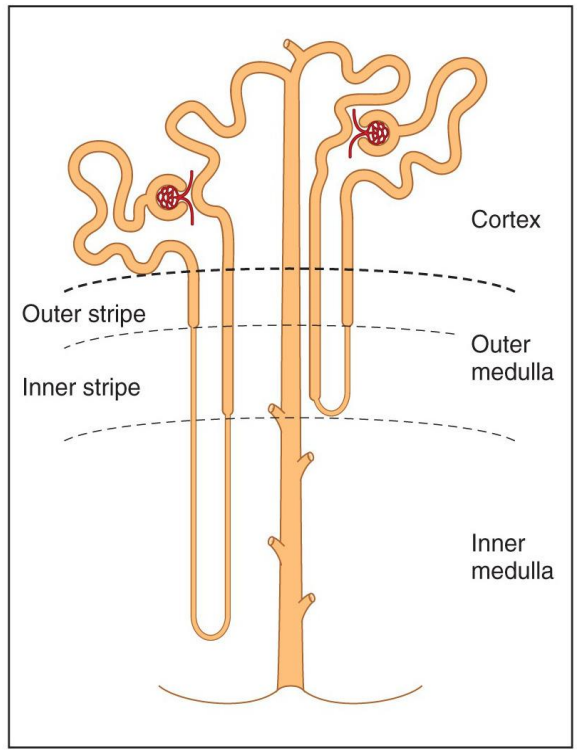
OVERVIEW OF KIDNEY FUNCTION

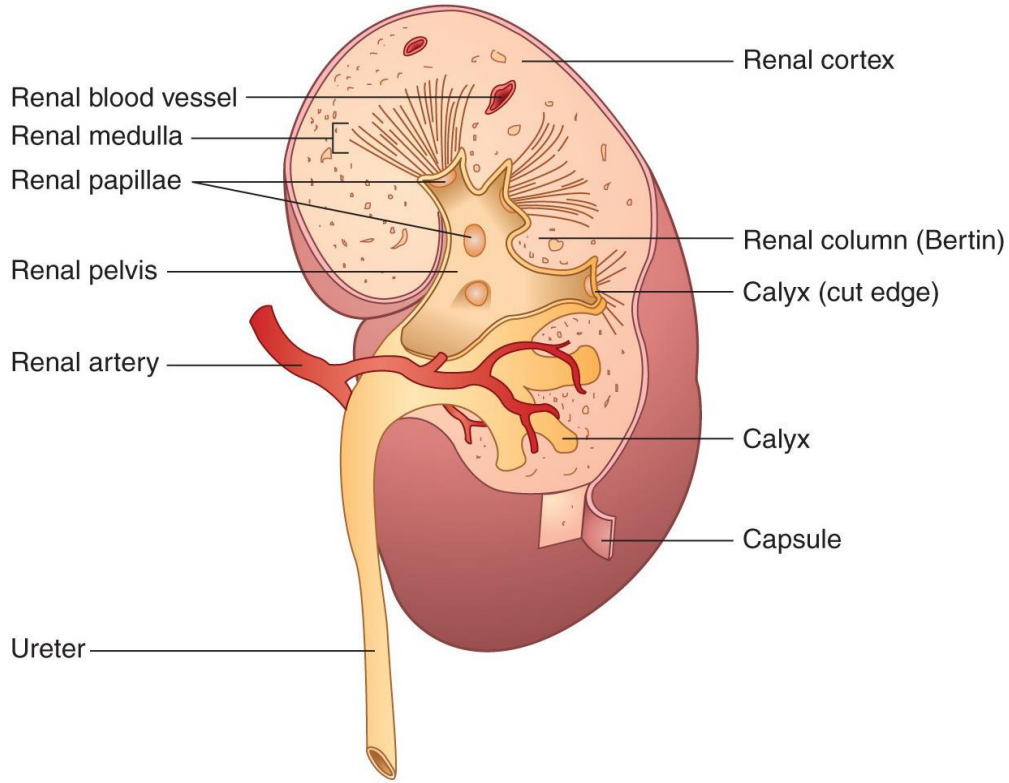
- Filtration of blood
- Regulation of fluid, electrolytes, and acid-base balance
- Endocrine functions (EPO, vitamin D activation, RAAS)



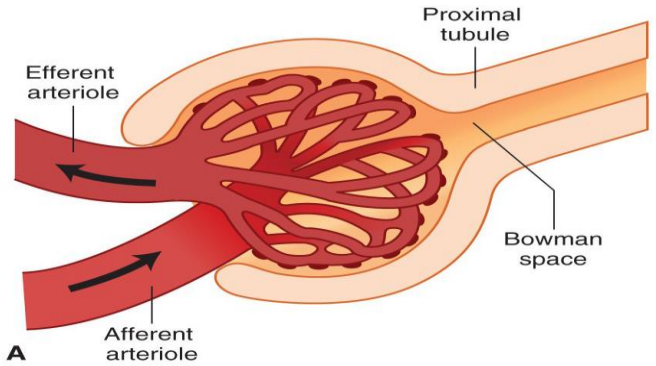


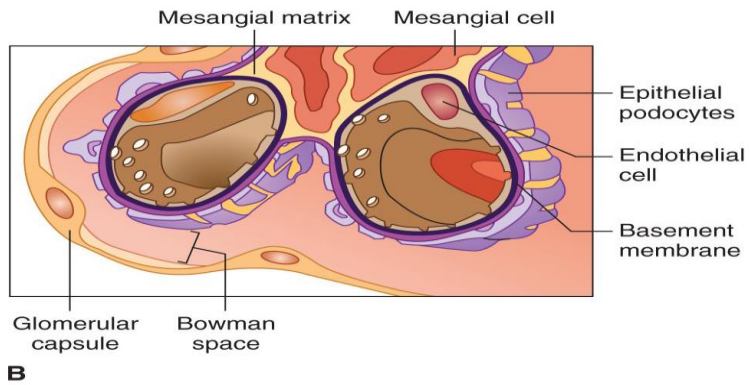
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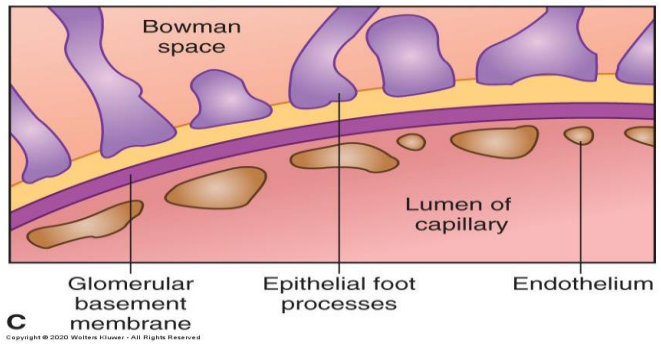




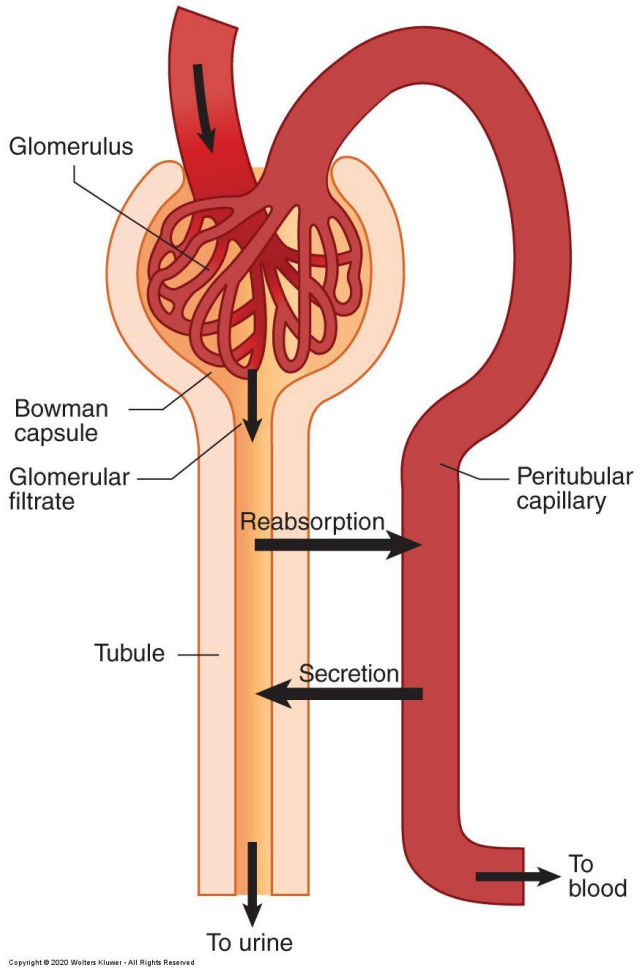
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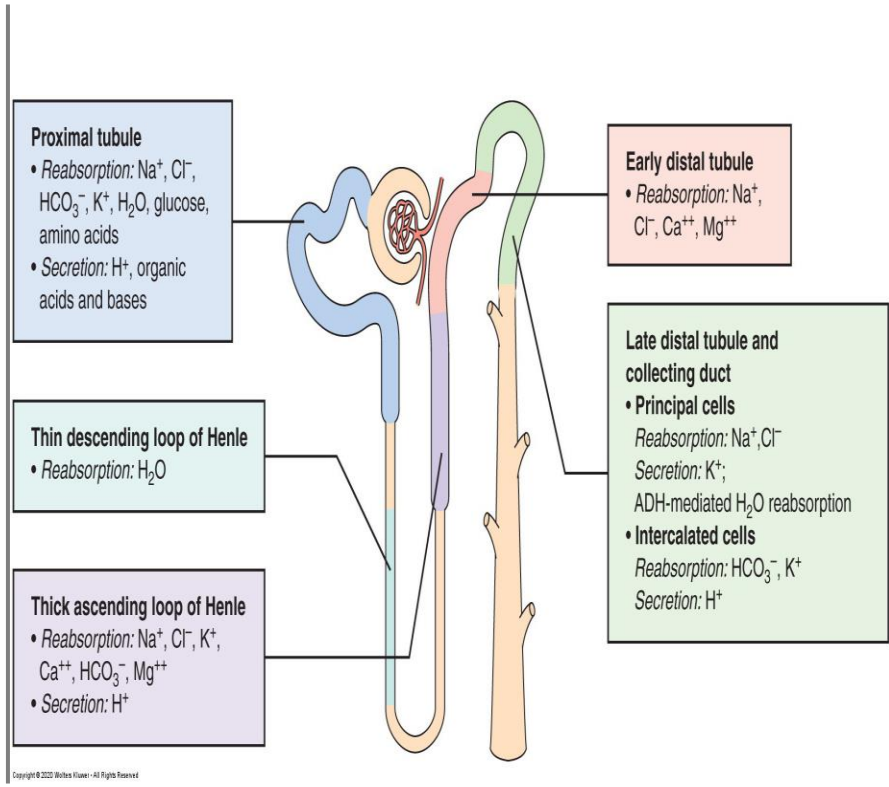




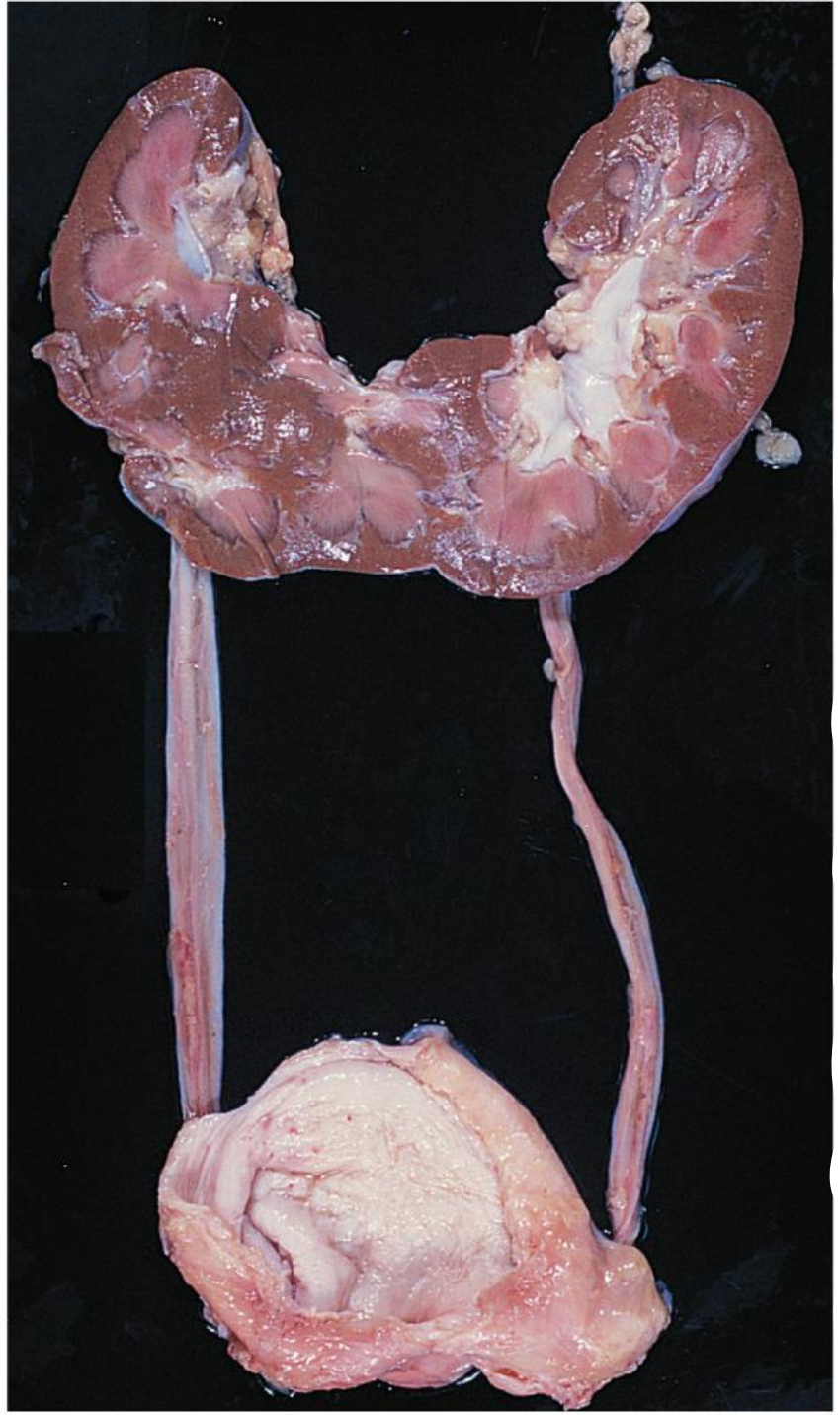
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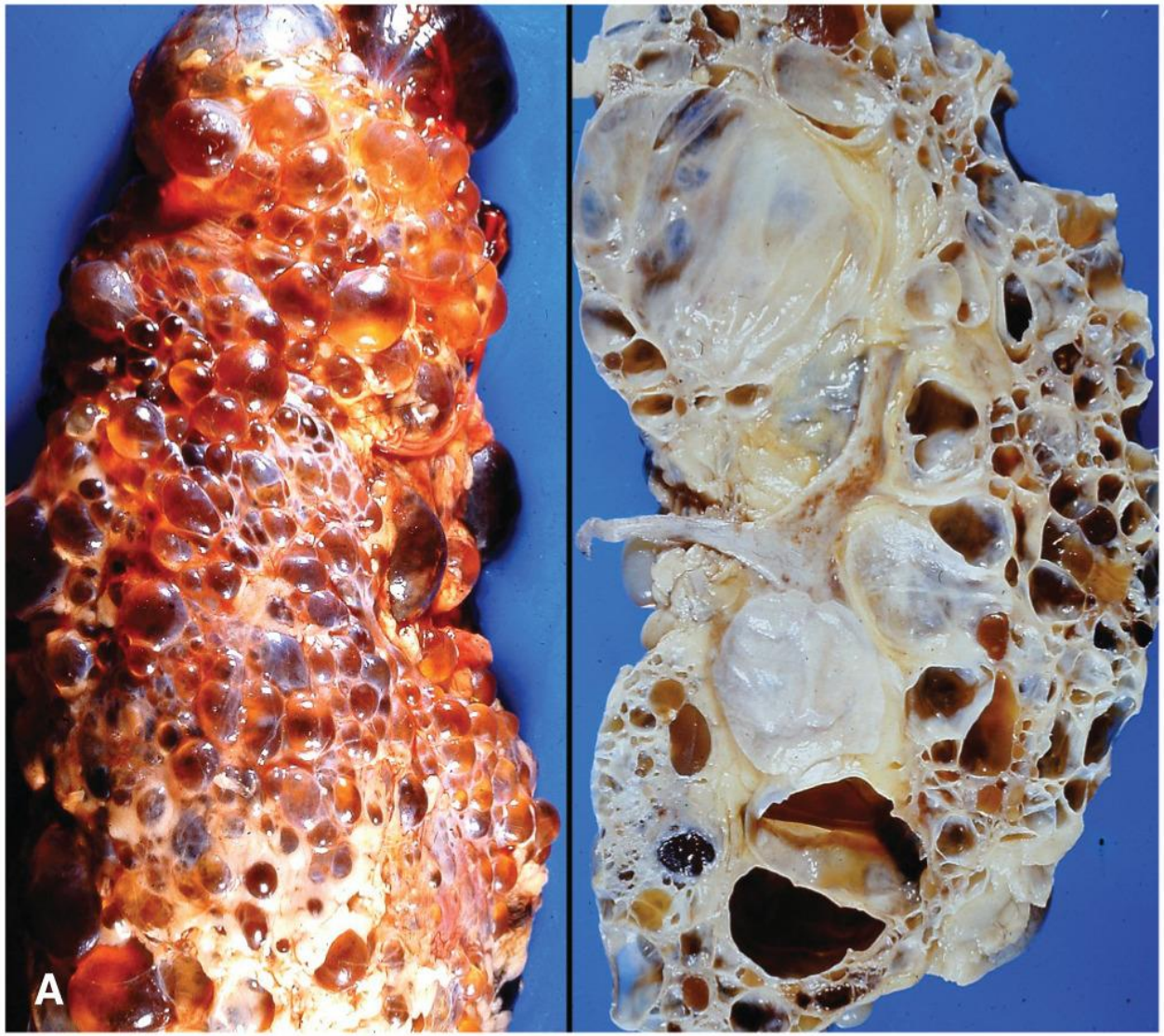


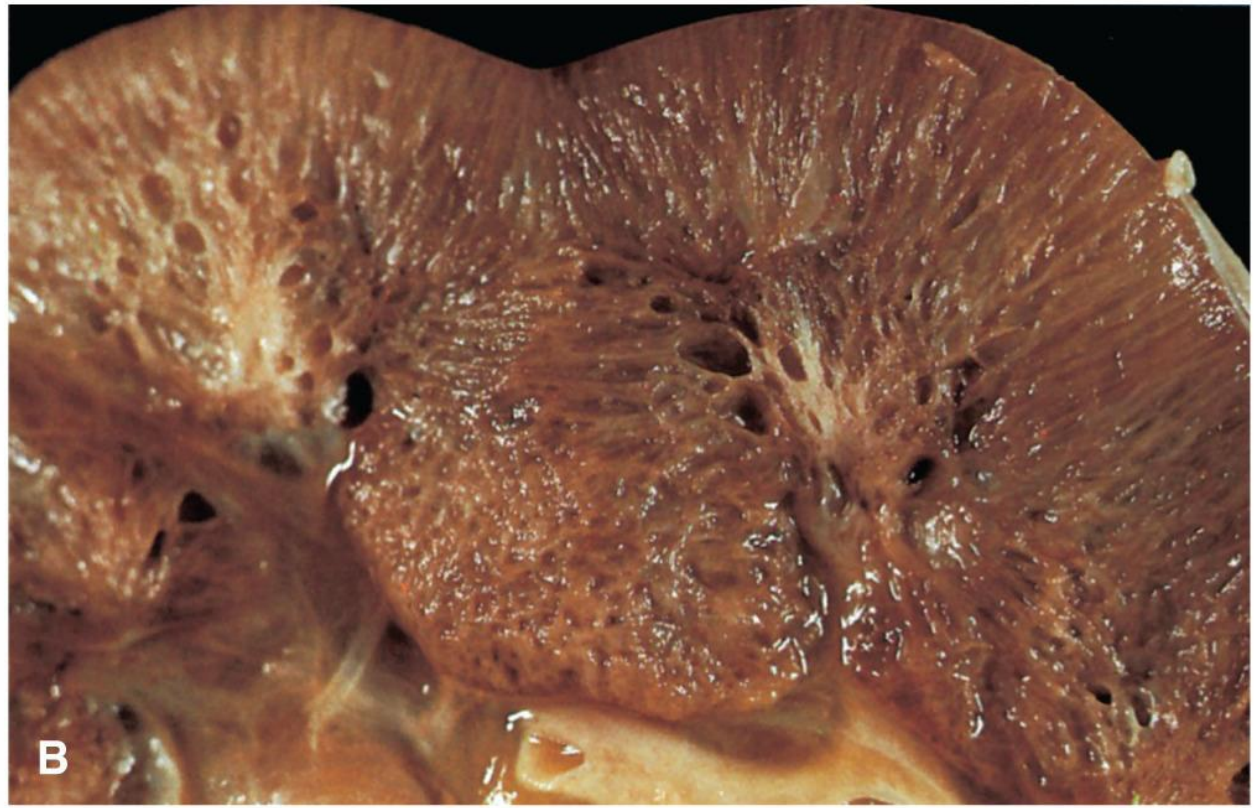
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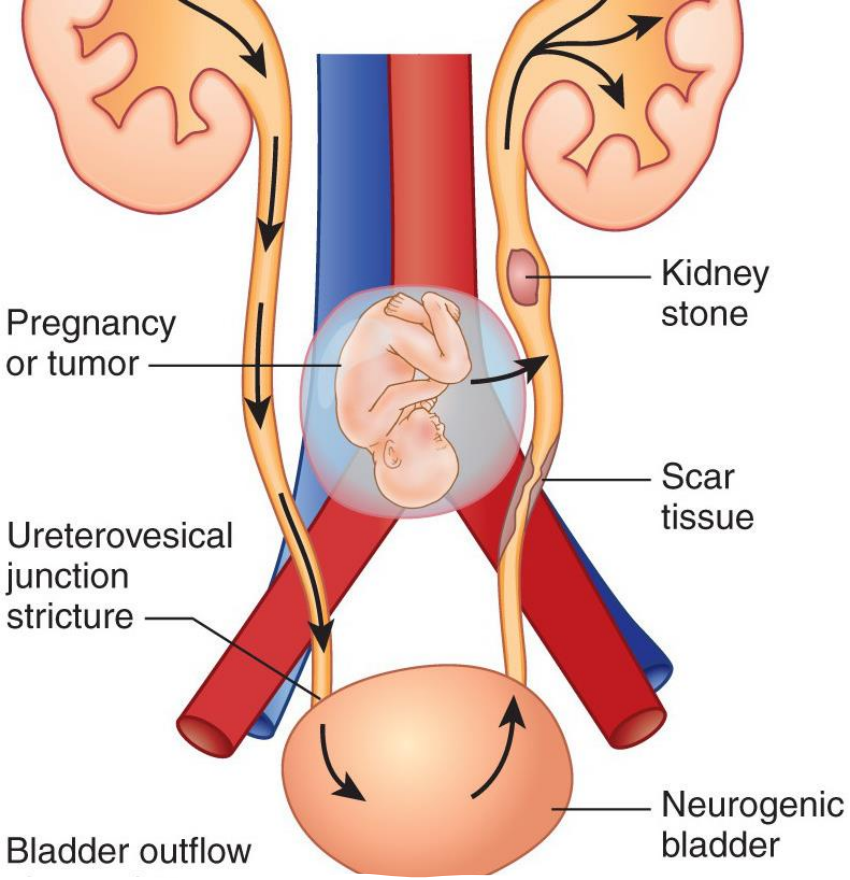
Developmental Defects of the Renal System

- **Definition:** Congenital anomalies of the kidney and urinary tract .
- **Common Defects:**
  - **Renal Agenesis** (Unilateral/Bilateral) → Potter syndrome (fatal in bilateral cases).
  - **Horseshoe Kidney** → Fusion of lower kidney poles.
  - **Polycystic Kidney Disease (PKD)** → Autosomal dominant vs. recessive forms.
  - **Vesicoureteral Reflux (VUR)** → Urine backflow from bladder to ureters





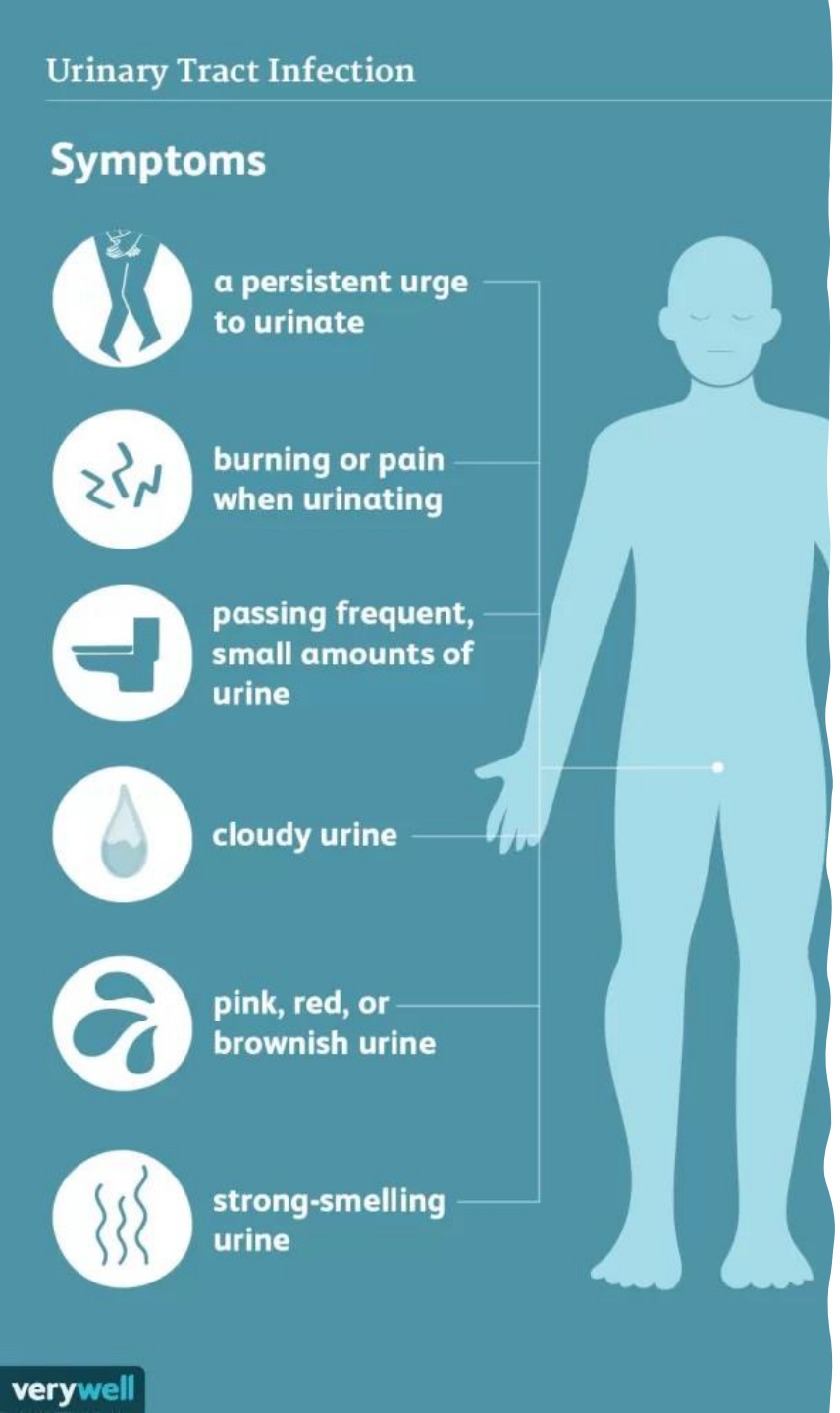
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Urinary Tract Obstruction

- **Stasis of urine**
  - Predisposes to infection and stone formation
- **Development of backpressure**
  - Interferes with renal blood flow and destroys kidney tissue
- **Common Symptoms:**
  - Pain
  - Signs and symptoms of UTI
  - Manifestations of renal dysfunction



Urinary Tract Infections (UTIs)

- **Definition:** Infection of the urinary tract (bladder, ureters, kidney).
- **Most common cause:** *Escherichia coli (E. coli)* (~80% of cases).
- **Risk factors:**
  - Female anatomy (shorter urethra).
  - Catheterization.
  - Pregnancy.
  - Diabetes.
  - Poor hygiene.

Urinary Tract Infections (UTIs)

- **Pathophysiology:**
  - Bacterial colonization → Urethral entry.
  - **Cystitis:** Localized bladder infection (frequency, dysuria, urgency).
  - **Pyelonephritis:** Infection ascends to kidneys → inflammation, fever, flank pain.
  - Recurrent infections → Chronic pyelonephritis → Renal scarring → CKD.
- **Clinical Features & Management:**
  - Symptoms: Dysuria, suprapubic pain, cloudy urine, hematuria, fever (pyelonephritis).
  - Diagnosis: Urinalysis, urine culture.
  - Treatment:
    - **Uncomplicated UTI:** Nitrofurantoin, Trimethoprim-Sulfamethoxazole.
    - **Pyelonephritis:** IV antibiotics (ceftriaxone, fluoroquinolones).
    - **Prevention:** Hydration, wiping front to back, cranberry supplement

Glomerulonephritis

is a group of kidney disorders characterized by inflammation of the glomeruli, the tiny filtering units of the kidneys. This inflammation disrupts kidney function, leading to altered filtration, protein loss, and, in some cases, kidney failure. GN can result from infections, autoimmune diseases, or systemic conditions affecting the kidneys

Nephrotic vs. Nephritic Syndrome

Feature	Nephrotic Syndrome	Nephritic Syndrome
Cause	Podocyte damage	Glomerular inflammation
Proteinuria	**>3.5 g/day** (Massive)	**Mild to moderate**
Hematuria	No	**Yes (RBC casts)**
Edema	Generalized (anasarca)	Mild periorbital edema
BP	Normal or low	**Hypertension**
Common Causes	Diabetes, Minimal Change Disease	Post-Strep Glomerulonephritis, Lupus

Pathophysiology

- **Pathophysiology:**
- **Nephrotic Syndrome:**
  - Damage to **podocytes** → Increased glomerular permeability.
  - **Massive protein loss** → Hypoalbuminemia → Edema.
  - **Hyperlipidemia** (compensatory liver response).
- **Nephritic Syndrome:**
  - Immune complex deposition (Post-strep, SLE) → **Glomerular inflammation**.
  - **Capillary damage** → Hematuria (RBC casts), mild proteinuria.
  - **Inflammation** → Increased BP, reduced GFR → Oliguria



Renal Stones (Nephrolithiasis)

- **Definition:** Hard mineral deposits in the renal system.
- **Types:**
  - Calcium oxalate (most common).
  - Uric acid (seen in gout).
  - Struvite (infection-related).
  - Cystine (genetic disorder).

Renal Stones

- **Pathophysiology:**
  - Supersaturation of urine with minerals.
  - Nucleation → Crystal formation.
  - Stone growth → Migration → Obstruction.
- **Clinical Features & Management:**
  - Symptoms: Severe flank pain (renal
- colic), hematuria, nausea/vomiting.
  - Diagnosis: CT scan (gold standard), urinalysis.
  - Treatment:
    - Hydration, pain control (NSAIDs, opioids).
    - Lithotripsy for large stones.
    - Surgical removal if obstructive

Neoplasms of the Kidney

A. Benign & Malignant Tumors

1. **Renal Cell Carcinoma (RCC)** – Most common adult kidney cancer
2. **Wilms Tumor (Nephroblastoma)** – Pediatric renal cancer
3. **Angiomyolipoma** – Benign renal tumor

- **B. Pathophysiology of Renal Cell Carcinoma:**
- Genetic mutations (VHL gene) → uncontrolled cell growth → local invasion & metastasis
- **C. Clinical Manifestations:**
- Hematuria, flank pain, palpable mass, weight loss

Discussion Questions:

1. Why are females more prone to UTIs than males?
2. What are the key differences between nephrotic and nephritic syndrome?

- Sarah, a 16-year-old female, presents to the emergency department with complaints of facial puffiness, swelling in her legs, and tea-colored urine for the past three days. She also reports mild headache, fatigue, and a decrease in urine output. Two weeks prior, she had a sore throat, which resolved without antibiotic treatment.
- Upon examination, the nurse notes:
- **Vital Signs:** BP 150/95 mmHg, HR 90 bpm, RR 18 breaths/min
- **Physical Exam Findings:**
  - **Periorbital edema and lower limb edema**
  - **Oliguria (low urine output)**
  - **Tea-colored (hematuric) urine**
  - **Mild tenderness in the flank area**

- **Laboratory Results:**
- **Urinalysis:**
  - Hematuria (+++)
  - Proteinuria (++)
  - RBC casts present
- **Serum Creatinine:** 2.0 mg/dL (↑)
- **Blood Urea Nitrogen (BUN):** 40 mg/dL (↑)
- **Serum Complement (C3):** Low
- **ASO (Anti-Streptolysin O) Titer:** Elevated
- The physician diagnoses **Post-Streptococcal Glomerulonephritis (PSGN)** and admits Sarah for close monitoring of kidney function and blood pressure. The nurse is responsible for monitoring her urine output, managing her hypertension, providing fluid and dietary guidance, and educating her and her family about the disease.

Discussion Questions:

1. What is the pathophysiology of post-streptococcal glomerulonephritis?
2. What are the nursing priorities in managing Sarah's condition?
3. Why is Sarah experiencing high blood pressure, and how should it be managed?
4. What dietary restrictions should be recommended?
5. How can nurses educate Sarah and her family about preventing recurrent infections and kidney damage?